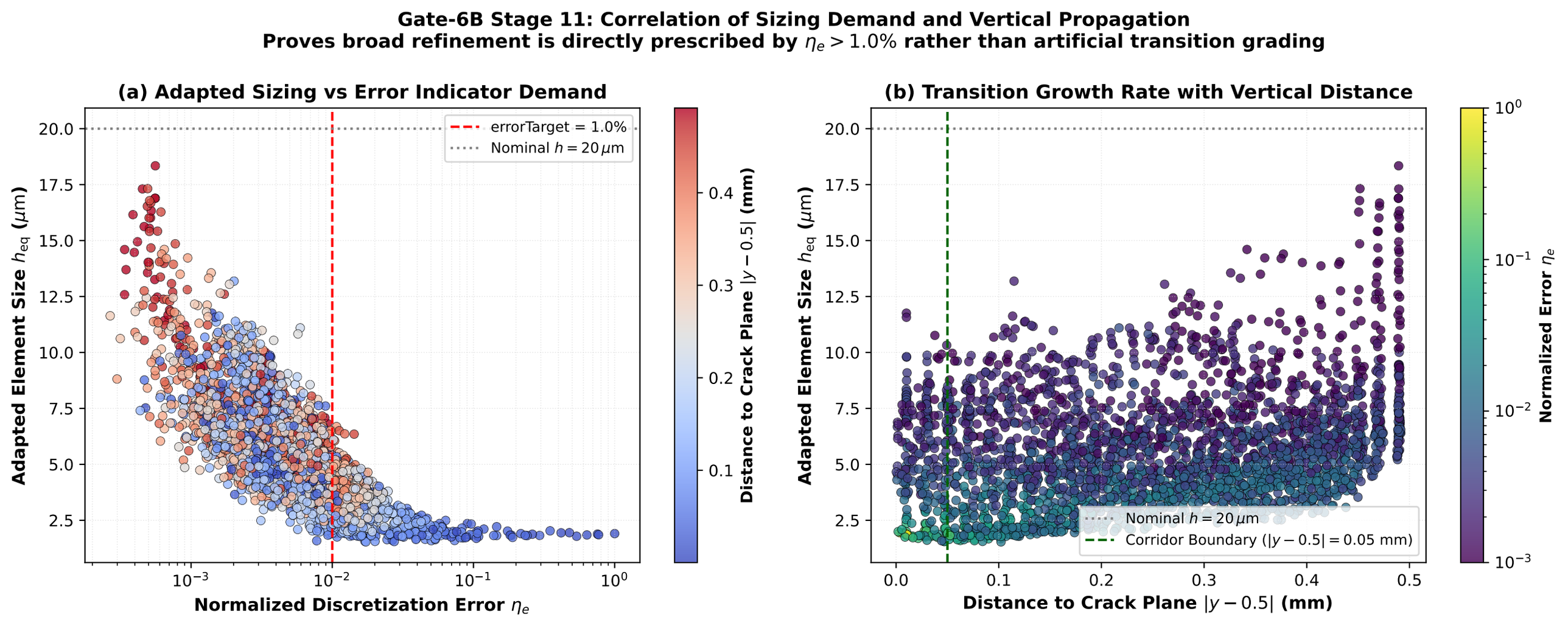

Gate-6B Stage 11: Correlation of Sizing Demand and Vertical Propagation
Proves broad refinement is directly prescribed by $\eta_e > 1.0\%$ rather than artificial transition grading
(a) Adapted Sizing vs Error Indicator Demand
errorTarget = 1.0%
Nominal $h = 20\,\mu$m
Adapted Element Size h_{eq} (μm)
Normalized Discretization Error η_e
Distance to Crack Plane $|y - 0.5|$ (mm)
(b) Transition Growth Rate with Vertical Distance
Nominal $h = 20\,\mu$m
Corridor Boundary ($|y - 0.5| = 0.05$ mm)
Adapted Element Size h_{eq} (μm)
Distance to Crack Plane $|y - 0.5|$ (mm)
Normalized Error η_e